REVIEW SUMMARY

SOLAR CELLS

# Rethinking the A cation in halide perovskites

Jin-Wook Lee†, Shaun Tan†, Sang Il Seok*, Yang Yang*, Nam-Gyu Park*

**BACKGROUND:** Organic-inorganic lead halide perovskites (OLHPs) represent a family of semiconducting materials that have attracted wide interest for their optoelectronic applications. By combining notable electronic and photophysical properties with low-temperature solution processability using earth-abundant materials, the field has experienced rapid breakthroughs over the past decade. OLHP-based optoelectronic devices have now been established as an independent field of study. In particular, solar cells that use OLHPs as the photoactive layer have exceeded 25% power conversion efficiency, which is competitive with photovoltaics based on traditional inorganic semiconductors such as silicon. However, unanswered questions remain to be addressed regarding the fundamental properties and our understanding of OLHP materials.

**ADVANCES:** OLHPs have the general chemical formula of $ABX_3$, where A represents a monovalent cation, B represents the divalent lead cation, and X is a halide anion. During its initial development, the A cation was limited to mainly three candidates, namely methylammonium ($MA^+$), formamidinium ($FA^+$), and cesium ($Cs^+$), which were determined by size and structural constraints to fit within the spaces of the OLHP crystal lattice. Because the A cation, by itself, does not directly contribute to the OLHP band-edge construction, it was traditionally believed to hardly affect the optoelectronic properties of OLHPs.

In contrast to these presumptions, advancements in recent years have unraveled the critical role of the A cation in determining the optoelectronic and physicochemical properties of OLHPs. Major breakthroughs in the field have been realized by using A cations to fine-tune the properties of OLHPs, including for (i) thermodynamic or kinetic stabilization of desirable OLHP phases, especially the metastable cubic formamidinium lead triiodide polymorph; (ii) impeding the migration of ions by electrostatic interactions and/or steric effects; and (iii) surface and interface functionalization and modification. These topics are now at the forefront of OLHP research, ultimately dictating the utility, function, performance, and stability of OLHP-based optoelectronic devices.

In this review, we examine the important developments enabled by the expanded role of the A cation. To highlight its importance, the vast majority of the record-breaking performances for OLHP-based photovoltaics have been enabled by breakthroughs related to the versatility of the A cation, which we discuss. Moreover, recently, bulky conjugated A cations have even been demonstrated to participate in constructing the OLHP band-edge structure, further challenging a long-held traditional notion regarding the role of the A cation.

**OUTLOOK:** Future opportunities entail either discovering new uses for existing A cation species or identifying new A cations for existing applications, or both simultaneously. When looking toward improving the commercialization readiness of OLHPs, A cation applications that involve addressing the module scale-up challenges and long-term operational instability issues must be further explored. This necessitates a more comprehensive understanding of the structure-property-performance-stability correlations conferred by the A cation by combining systematic experimental approaches with first-principles theoretical studies. ■

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Cite this article as J.-W. Lee et al., *Science* **375**, eabj1186 (2022). DOI: 10.1126/science.abj1186

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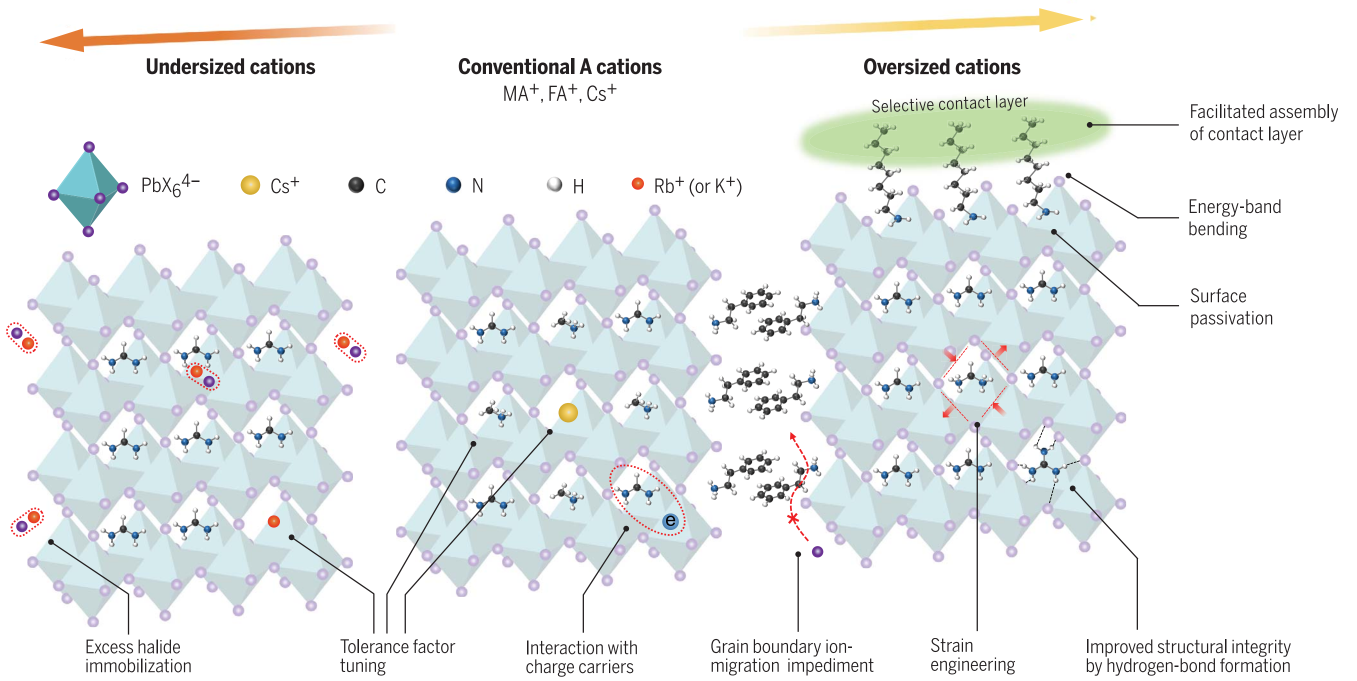
https://doi.org/10.1126/science.abj1186






**Modifying perovskites through A site cations.** In recent years, important breakthroughs have been enabled by the expanded role and function of the A cation in tailoring the physicochemical and optoelectronic properties of OLHPs. These advances ultimately dictate the utility, function, and performance of OLHP-based optoelectronic devices.

REVIEW

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# Rethinking the A cation in halide perovskites

Jin-Wook Lee[1]†, Shaun Tan[2,3]†, Sang Il Seok[4]*, Yang Yang[2,3]*, Nam-Gyu Park[5]*

The A cation in $ABX_3$ organic-inorganic lead halide perovskites (OLHPs) was conventionally believed to hardly affect their optoelectronic properties. However, more recent developments have unraveled the critical role of the A cation in the regulation of the physicochemical and optoelectronic properties of OLHPs. We review the important breakthroughs enabled by the versatility of the A cation and highlight potential opportunities and unanswered questions related to the A cation in OLHPs.

Organic-inorganic lead halide perovskites (OLHPs) have the general stoichiometry formula of $ABX_3$, with A as the monovalent organic or inorganic cation, B as the divalent $Pb^{2+}$ cation, and X as the halide anion. Any of the A, B, and X sites can be occupied by a mixture of constituents to afford OLHPs a wide compositional breadth and tuneability toward different applications (*1*, *2*). The Goldschmidt tolerance factor can be useful to predict structural formability and the resulting crystal phase based on ionic size considerations; it is defined by

$$t = \frac{r_A + r_X}{\sqrt{2}(r_B + r_X)}$$

where $r_A$, $r_B$, and $r_X$ are the ionic radii of the A, B, and X species, respectively. The perovskite phase is desired for many practical optoelectronic applications and empirically observed to form at $0.8 < t < 1.0$. The crystal structure consists of an inorganic $BX_6^{4-}$ octahedra network that is three-dimensionally connected through the corner halides, with structural stabilization contributed by the A cations that occupy the cuboctahedral spaces surrounded by eight $BX_6^{4-}$ octahedra. For $APbI_3$-based stoichiometries, three A cations are known to definitively fit this geometric constraint, namely methylammonium ($CH_3NH_3^+$, or $MA^+$), formamidinium [$HC(NH_2)_2^+$, or $FA^+$], and cesium ($Cs^+$), albeit $FAPbI_3$ ($t \sim 0.99$) and $CsPbI_3$ ($t \sim 0.80$) are at the edges of the $t$ limits.

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Traditionally, the A cation was not believed to directly contribute to the OLHP band-edge construction, which is primarily composed of orbital states from the B and X species (*3*). Resultantly, the A cation was not thought to substantially affect the optoelectronic properties of OLHPs. However, progress in recent years has since challenged this notion. Furthermore, the A cation has been shown to critically influence the physiochemical properties and the intrinsic and extrinsic stability of OLHPs.

In this review, we evaluate the expanded role of the A cation (Fig. 1), which has enabled many important breakthroughs for high-performance and stable OLHP optoelectronics. Many of these developments were made possible by the introduction of oversized ($t > 1$) or undersized ($t < 0.8$) cations that may not necessarily fit the structural constraints of the lattice bulk. Conversely, such missized cations may occupy the A sites along the surfaces and/or grain boundaries, which are free from the bulk steric constraints. By themselves, the oversized cations typically form lower-dimensional phases, such as the two-dimensional (2D) $A_2PbI_4$ perovskites, whereas the undersized cations are used to form the $ABO_3$ oxide perovskite structures. We focus only on their utility to and function in tuning the properties of three-dimensional (3D) OLHPs.

## Revisiting the conventional roles of the A cation: Progress and challenges

### *Crystal symmetries and crystal phases*

The size and geometry of the A cation affect the bond length and angle between the B and X species to alter the arrangement of the surrounding $BX_6^{4-}$ octahedra and thus the resulting crystal symmetry and phase of the perovskite. The molecular orbitals of the A cation constitute deep energy states within the conduction and valence bands and thus do not directly affect the band-edge carrier properties (*4*). Consequently, A cation engineering has been considered as a useful approach to fine-tune the crystal structure (orthorhombic, tetragonal, and cubic) and physicochemical properties of OLHPs without substantially altering their optoelectronic properties.

The organic A cations interact with the surrounding $BX_6^{4-}$ octahedra through weak secondary hydrogen bonding, with bonding energies lower than 0.1 eV per bond (*5*). Owing to such weak interaction energies, the reorientation of the organic cation can be thermally activated at a finite temperature. [The residence time at room temperature is on the order of ~$10^2$ femtoseconds to picoseconds (*6*–*8*).] The activated reorientation of the organic A cation and its coupling with the surrounding inorganic framework result in a temperature-dependent order-disorder–type phase transition between the orthorhombic, tetragonal, and cubic phases (*9*, *10*). For example, $MAPbBr_3$ crystallizes in the orthorhombic phase at low temperatures (<180 K) where the $MA^+$ cations are ordered. As the temperature increases to more than 180 K, the $MA^+$ cation becomes disordered with anisotropic thermal motion, which is coupled to the surrounding $PbBr_6^{4-}$ octahedra to induce the crystal phase transition to the tetragonal symmetry (Fig. 2A) (*11*). Contribution of the A cation thermal motion in relation to the OLHP phase behavior has also been discussed for $FAPbI_3$. The $FA^+$ cation, which has a larger ionic radius than $MA^+$, cannot be stabilized in the cubic symmetry at temperatures below ~390 K, and thus it forms a hexagonal nonperovskite polymorph. At temperatures beyond ~390 K, the activated thermal motion of the $FA^+$ cation contributes entropic stabilization to the cubic perovskite polymorph, which is subsequently retained as a metastable phase at temperatures lower than the phase conversion temperature (*12*). Because the thermal motion of the organic A cation and its coupling with the surrounding inorganic sublattice dictates the crystal structure of the OLHP system and its stability at operating temperatures, not only the static geometry of the A cation but also its dynamic structure should be considered in designing the A cation. In this regard, to predict structural formability, the effective ionic radii of the nonspherical organic cations have been estimated by assuming rotational freedom about the center of mass, by adding the distance from the center of mass to the outermost ion (excluding hydrogen) to the radius of the ion (*13*). However, the recalculated $t$ values for many organic cations are still not correlated with experimental results (*14*). Introducing other geometrical factors such as globularity improved the correlation, but the versatility of these approaches has not yet been confirmed (*14*). Further work is needed to enhance the accuracy of calculating $t$, which may include additional corrections to account for molecular symmetry and hydrogen-bond strength, as well as the A cation dipole.

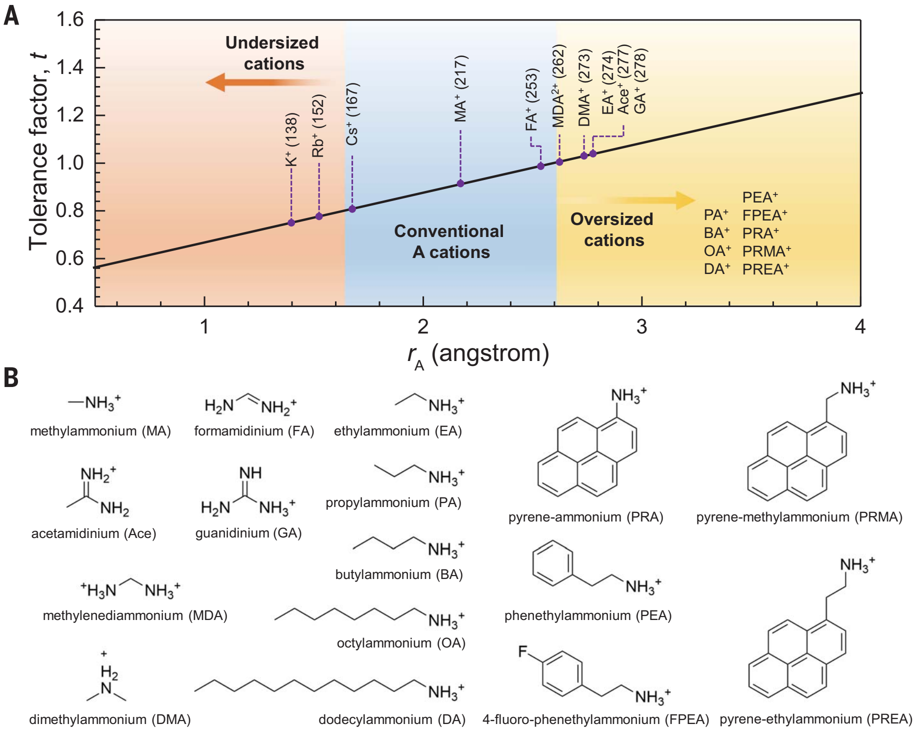


**Fig. 1. Expanded role of A cations.** (**A**) Selection of A cations that have enabled important breakthroughs in the field of OLHPs, classified by their cationic radius and corresponding tolerance factor based on the $APbI_3$ lattice framework. The number in parentheses indicates the cation radius. (**B**) Molecular structures of the organic cations listed in (A).

### *Compositional breadth and tunability for bandgap engineering*

State-of-the-art, high-performance single-junction perovskite solar cells (PSCs) primarily incorporate $Pb^{2+}$ and $I^-$ in their B and X sites, respectively, because of the desirable bandgaps of such stoichiometries, which are closer to the ideal Shockley-Queisser theoretical value (~1.34 eV). Conversely, wide-bandgap OLHPs (~1.7 eV) are actively investigated for fabricating the top subcell of perovskite-based tandem solar cells by halide compositional tuning (*15*). However, continuous halide compositional tuning is not achievable for $FAPb(I_{1-x}Br_x)_3$ because the $t$ value of $FAPbI_3$ is close to the cubic phase limit, and Br incorporation further increases $t$ and reduces the A cuboctahedral space to aggravate structural formability. This results in the formation of an amorphous phase with high energetic disorder, low absorption, and inferior carrier mobility (~1 $cm^2\ V^{-1}$) compared with the pure iodide phase (~20 $cm^2\ V^{-1}$) (*16*, *17*). By contrast, A cation compositional engineering offers a solution to enable access to the wider bandgap range. Partial replacement of $FA^+$ with the smaller $MA^+$ or $Cs^+$ cation reduces the $t$ value to compensate for the increase in $t$ owing to the Br incorporation (*18*, *19*). This enables the formation of crystalline wide-bandgap compositions such as $FA_{0.83}Cs_{0.17}Pb(Br_{0.4}I_{0.6})_3$, with an optical bandgap of ~1.74 eV and mobility of 21 $cm^2\ V^{-1}$. Such multication mixed-halide compositions are now widely used for fabricating high-efficiency perovskite-silicon or perovskite-perovskite tandems (*20*–*22*).

Although widely used, the multication mixed-halide compositions suffer from spontaneous photo-induced phase segregation (*23*). The compositional engineering of A cations offers an advantage in this regard. In addition to compensating for $t$ changes by halide substitution as discussed, more recently, A cation compositional engineering has been used to directly modulate the bandgap of the perovskite without changing the halide composition. $CsPbI_3$ has a suitable bandgap of ~1.7 eV, but its cubic phase is thermodynamically unstable at room temperature because of its low $t$ value. Additive engineering has enabled the synthesis of thermodynamically stabilized $CsPbI_3$ with a power conversion efficiency (PCE) exceeding 19%, where $CsPbI_3$ is synthesized using dimethylammonium iodide (DMAI) or $HPbI_3$ (*24*, *25*). Quasi-2D perovskites that incorporate oversized ($t$ > 1) ammonium cations are another class of wide-bandgap OLHPs that have been broadly explored (*26*). The bulky insulating ammonium cations that are intercalated between the 3D OLHP layers induce charge-carrier quantum confinement to enable bandgap engineering by varying the number of 3D layers (Fig. 2B) (*27*). However, the anisotropic charge transport and poor phase purity have limited the performance of such PSCs (*28*). Further developments are necessary for effective methods to grow phase-pure quasi-2D OLHPs with a desired orientation, because they might be promising active material candidates for the top subcell of perovskite-based tandems owing to their superior stability compared with conventional 3D OLHPs.

### *Charge-carrier dynamics*

Although band-edge carrier transport is mediated through the $BX_6^{4-}$ network, the thermally activated reorientational motion of dipolar A cations and coupled $BX_6^{4-}$ octahedra may interact with charge carriers to affect their dynamics (*7*). It was suggested that the formation of (ferroelectric) large polarons, that is, charge carriers dressed by long-range lattice deformations (Fig. 2C), may be the possible origin of the long carrier lifetimes of OLHPs (*29*, *30*). The hot-carrier lifetime in OLHPs was found to be ~$10^2$ ps, which is exceptionally long compared with the ~$10^2$ fs lifetime of conventional polar semiconductors (*7*, *31*). This further suggests the potential application of OLHPs to hot-carrier solar cells, which remain relatively unexplored. It was proposed that the ultrafast reorientational motion of the A cation in response to the photogeneration of charge carriers may form large polarons, resulting in an effective screening of the Coulomb potential to diminish hot-carrier scattering (*7*). The long-lived hot carriers are observed in $MAPbBr_3$ and $FAPbBr_3$, but not $CsPbBr_3$. However, separate studies show an opposite trend, where the slower hot-carrier cooling of $CsPbBr_3$ is attributed to a relatively weak charge carrier–phonon coupling compared with its hybrid counterparts (*32*). Nevertheless, these studies highlight the importance of the dynamic motion of the A cation.

The role of the A cation in the band-edge carrier dynamics is still under debate. In contrast to initial reports correlating polaronic stabilization with A cations (*7*), recent consensus seems to be that the generation of large polarons is more closely associated with the dipolar $BX_6^{4-}$ sublattice rather than the A cation, which is supported by the similar radiative recombination kinetics of band-edge carriers in $MAPbBr_3$, $FAPbBr_3$, and $CsPbBr_3$ (*33*). Conversely, experimental observations still indicate that the lifetimes of band-edge carriers are greatly affected by the A cation composition. For example, $FAPbI_3$-based compositions have demonstrated much longer charge-carrier lifetimes compared with that of $MAPbI_3$, irrespective of the former's generally inferior crystallinity and poor phase purity (*34*, *35*). Based on solid-state nuclear magnetic resonance (NMR) investigations, there exists a correlation between the faster reorientation

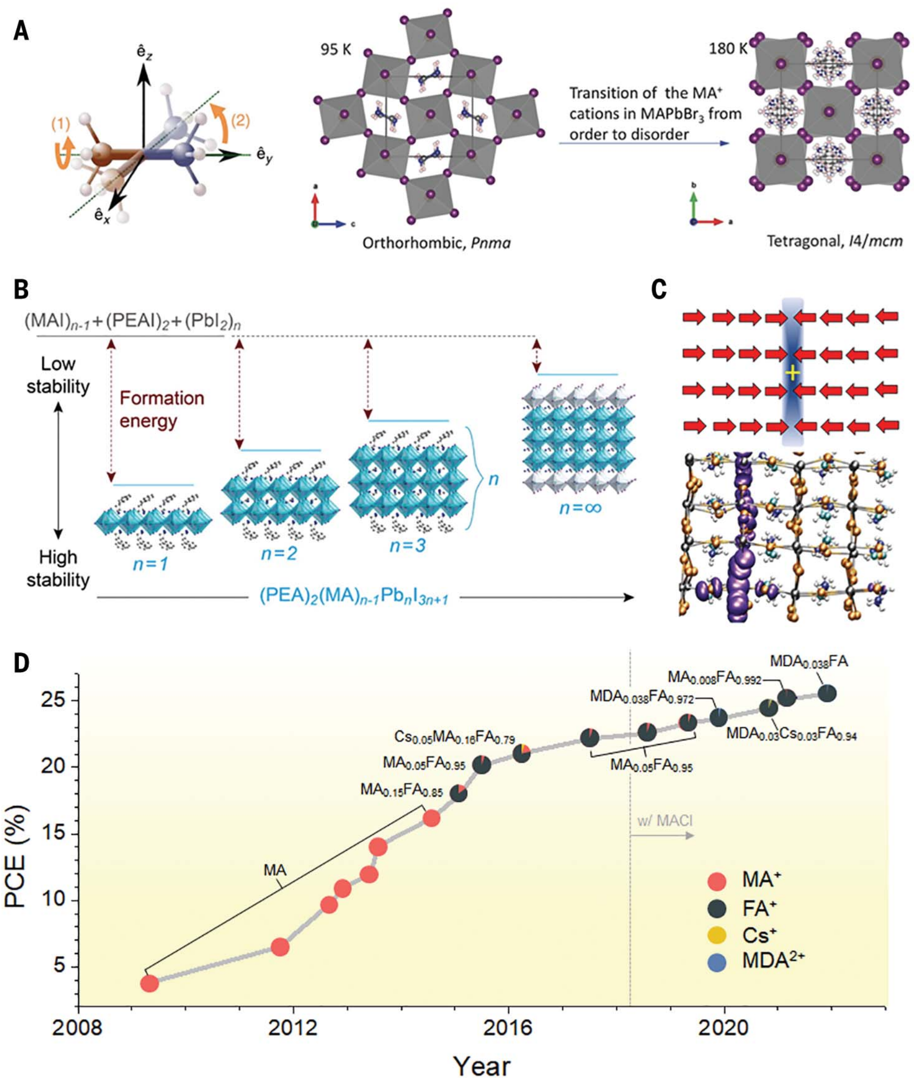


**Fig. 2. Revisiting the conventional roles of the A cation.** (**A**) Schematics illustrating the dynamic motion of the $MA^+$ cation (left) and temperature-dependent order-disorder–type phase transition of $MAPbBr_3$ induced by the $MA^+$ cation motion (right). Adapted with permission from (*6*) and (*11*). Labels (1) and (2) on the left figure indicate the two reorientation modes: (1) methyl and/or ammonium rotation around the C–N axis (carbon and nitrogen atoms are depicted in blue and brown, respectively, and the axis is indicated by the green dotted lines); and (2) whole-molecule reorientation via rotation of the C–N axis itself. (**B**) Low-dimensional $PEA_2MA_{n-1}Pb_nI_{3n+1}$ perovskites and their corresponding relative formation energy and stability. Adapted with permissions from (*27*). (**C**) Schematic of the ferroelectric large polarons and molecular dynamics simulation of the corresponding isodensity representation of holes, showing sheet-like charge localization in tetragonal $MAPbI_3$. Adapted with permission from (*29*). (**D**) History of the record PSC performance and the A cation compositions that were used. The PCE data and compositions are retrieved from their respective publications (*1*, *2*, *19*, *45*, *49*–*51*, *87*, *102*–*110*). The PCEs are plotted based on publication date. This panel conveniently summarizes the composition evolution of state-of-the-art PSCs with time, but we note that the final PSC efficiency is not solely determined by composition alone.

dynamics of the A cation and longer charge-carrier lifetimes (*36*). The temperature and resulting phase-dependent charge-carrier lifetime have also been correlated with the rotational entropy of the A cation (*37*). Recent computational studies proposed that charge carriers are weakly localized by the fluctuating organic A cation, although the subsequent energetic stabilization to form polarons is mediated by the $BX_6^{4-}$ sublattice (*38*). It was also suggested that polaron hopping is related to the random reorientation of the organic cations. This again implies that the dynamic structure of the organic A cation plays a crucial role in affecting OLHP optoelectronic properties. However, all the observed carrier dynamics could have also manifested from composition-dependent defect type, density, and energetics. Therefore, further studies are required to unravel the correlations between the A cation and charge-carrier dynamics, which are essential for designing OLHP compositions.

## Expanded role of the A cation

### *Phase stabilization of desired perovskite polymorphs*

More recent record-breaking PSCs contain continuously increasing proportions of $FA^+$ in their compositions (Fig. 2D). Compared with the prototypical $MAPbI_3$, $FAPbI_3$-dominant compositions generally boast longer carrier lifetimes, superior thermal and photostability, and a bandgap closer to the ideal for photovoltaic applications (*3*). Stabilization of the metastable cubic $FAPbI_3$ phase at room temperature has become an important focus of ongoing research in the community (Fig. 3A).

Initially, substitution with the smaller $MA^+$ cation was widely used to lower the $t$ value of $FAPbI_3$. Replacement of FAI with MAI or MABr in the precursor solution contributes to cubic phase stabilization at room temperature (*19*, *39*). The $(FAPbI_3)_{1-x}(MAPbI_3)_x$ or $(FAPbI_3)_{1-x}(MAPbBr_3)_x$ compositions demonstrated superior optoelectronic properties and stability compared with pure $MAPbI_3$ or $FAPbI_3$, resulting in several record-breaking certified efficiencies around 2015 (Fig. 2D) (*3*). However, $MA^+$ forms a smaller number of hydrogen bonds than $FA^+$ with the surrounding $PbI_6^{4-}$ inorganic framework, and its consequent volatile nature aggravates the thermal and photo instability of the OLHP films (*40*, *41*). This motivated the development of alternative cations to stabilize the cubic polymorph. The inorganic $Cs^+$ and $Rb^+$ cations have a smaller ionic radius than $FA^+$ but form stronger primary chemical bonding with the $BX_6^{4-}$ framework. Consequently, $Cs^+$ and $Rb^+$ effectively stabilize the cubic phase at lower temperatures while simultaneously enhancing the environmental stability of the films (*41*–*43*). It was also suggested that A cation mixing contributes entropic stabilization to the system (*44*), which motivated the development of more complex triple or quadruple cation systems such as $FA_{1-x-y}MA_xCs_y$- or $FA_{1-x-y-z}MA_xCs_yRb_z$-based compositions (*45*, *46*). Regardless of the improved phase stability of most $FAPbI_3$-based compositions by substitution with $MA^+$, $Br^-$, or $Cs^+$, however, these approaches inevitably increase the bandgap to sacrifice the photocurrent.

Recently, the focus of the community has shifted to phase-stabilization strategies that can preserve the inherent bandgap of cubic $FAPbI_3$. To minimize the bandgap penalty, "additive removal" approaches have been developed by using the smaller $MA^+$ cation to first form a highly crystalline perovskite phase and then removing it afterward. MACl is, at present, the most widely used additive among

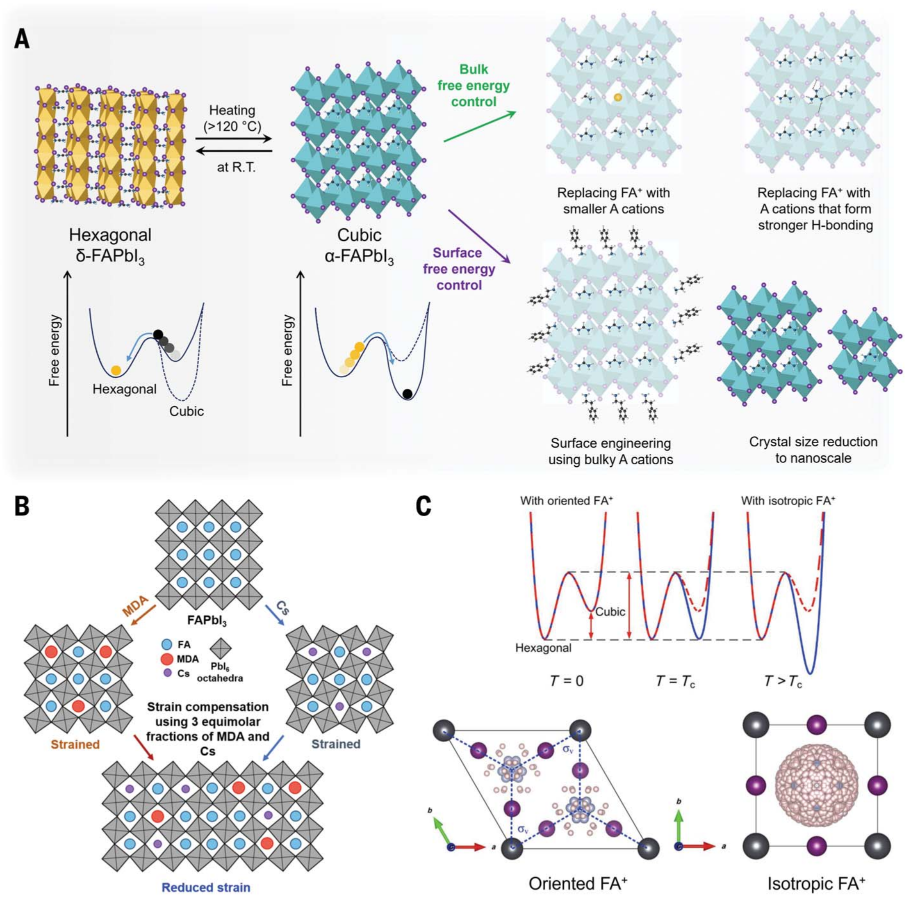


**Fig. 3. Stabilization of the cubic $FAPbI_3$ phase by A cation engineering.** (**A**) Schematics illustrating the metastability of the cubic $\alpha$-$FAPbI_3$ phase and its corresponding free-energy diagram (left) and approaches to stabilize the cubic $\alpha$-$FAPbI_3$ phase by tuning the bulk or surface free energy (right). R.T., room temperature. (**B**) Schematic illustration of lattice strain engineering by using $MDA^{2+}$ and $Cs^+$ cations. Adapted with permission from (*51*). (**C**) Schematics showing the free-energy diagrams at different temperatures and refined structures of $FAPbI_3$ at the corresponding temperatures. At temperatures lower than the phase-transition temperature ($T_c$), the $FA^+$ cation is oriented within the hexagonal crystal structure, whereas its orientation becomes isotropic (random) in the cubic crystal structure at temperatures higher than $T_c$, contributing to the entropy of the system. Adapted with permission from (*12*).

these approaches (*47*, *48*). The $MA^+$ cation from the added MACl initially participates in forming the perovskite intermediate phase at room temperature. During the subsequent annealing process, the intermediate phase is converted into the perovskite crystals with the vaporization of MACl. The slow vaporization of MACl and thus retarded crystal growth kinetics promote the growth of highly crystalline perovskite crystals. Regardless of the high molar concentrations of the excess MACl (~40 mol %) in the precursor solution, the final OLHP films contain only a small amount of residual $MA^+$ (~5 mol %) (*49*), and thus their bandgaps (~1.50 to 1.53 eV) are comparable to that of pure $FAPbI_3$. This approach enables the formation of phase-pure and highly crystalline $FAPbI_3$-based films, and it has been broadly used in studies that report record-efficiency PSCs (vertical dashed line in Fig. 2D) (*50*, *51*). Relatedly, the dual combination of MACl and CsCl was also found to effectively stabilize the perovskite phase (*52*). A similar approach was developed by using gaseous methylammonium thiocyanate (MASCN) (*53*). It was reported that MASCN is incorporated onto the surface of hexagonal $FAPbI_3$ crystals to assist in the conversion of the hexagonal phase into the cubic polymorph. For either MACl or MASCN, the counter anions were also reported to contribute to stabilizing the $FAPbI_3$ perovskite phase. Further developments may elucidate the different roles of the counter anions as well as explore alternative cations or counter anions that can impart additional functionalities.

The phase energetics of materials is governed by the Gibbs free energy of the phase, which is in turn determined by the summation of the bulk and surface free energies. Therefore, the phase conversion of metastable OLHP phases can be modulated by surface-energy manipulation. For example, it was demonstrated that the metastable $CsPbI_3$ perovskite phase can be stabilized by synthesizing nanosized crystals where the surface-energy contribution becomes dominant compared with that of its bulk counterpart (*54*). Such an approach was found to be also valid for $FAPbI_3$, where nanocrystalline $FAPbI_3$ demonstrated a superior phase stability compared with its corresponding bulk crystals (*55*). Surface-energy modulation can also be achieved by adopting surface-functionalizing oversized A cations (*56*–*58*). Bulky ammonium cations, such as phenethylammonium ($PEA^+$), butylammonium ($BA^+$), or isopropylammonium, are spontaneously expelled to the grain boundaries and surfaces during perovskite crystallization because of their inability to incorporate into the crystalline lattice bulk. The bulky cations are free from steric constraints at the surface and thus can coordinate with the surface A site by bonding with the surface $BX_6^{4-}$. Such surface functionalization was suggested to modulate the crystal surface energy to stabilize the cubic perovskite phase. This approach allows for the stabilization of metastable OLHP phases without changing its bulk composition and inherent bandgap.

The phase energetics of OLHPs were also suggested to be dependent on the hydrogen-bond number and bond strength between the A cation and surrounding inorganic framework (*5*). This is illustrated by using the divalent methylenediammonium ($MDA^{2+}$) cation that can form a greater number of hydrogen bonds with the $BX_6^{4-}$ lattice than $FA^+$ (*50*). Compared with $MA^+$, lower concentrations of $MDA^{2+}$ are required to stabilize the cubic phase of $FAPbI_3$, even though the ionic radius of $MDA^{2+}$ is slightly oversized compared with that of $FA^+$. The incorporation of $MDACl_2$ may generate FA vacancy and interstitial Cl defects to maintain charge neutrality but, importantly, without forming deep trap states. It is noteworthy that the PSC performance and stability improved regardless of the generated additional defects, but further investigations may be necessary to decouple their full consequences. Nevertheless, incorporation of $MDA^{2+}$ inadvertently induces tensile lattice strain as a result of its large cation radius. A direct correlation between residual lattice strain and a detrimental effect on the OLHP charge carrier dynamics has been reported (*59*). To compensate for the residual strain, it was further reported that the smaller $Cs^+$ cation can be incorporated in equimolar amounts with $MDA^{2+}$ (Fig. 3B) (*51*). Tensile strain relaxation by $Cs^+$ incorporation resultingly increased the carrier lifetime owing to a reduced defect concentration in the film. The case of $MDA^{2+}$ illustrates the possibility of expanding into higher-valency cations for use in OLHPs, although their exact phase-stabilization mechanisms still need to be explored.

More generally, the vast majority of $FAPbI_3$ stabilization strategies now reported could be considered kinetic stabilization approaches, such that the cubic $FAPbI_3$ phase only remains metastable and still spontaneously degrades into the equilibrium hexagonal phase. Although kinetic stabilization is effective in retarding phase degradation, true thermodynamic stabilization of cubic $FAPbI_3$ as the equilibrium phase remains the elusive goal. To achieve this, understanding the many overlapping factors that affect the thermodynamic energetic landscape of OLHPs is required. For example, the thermally activated random motion of the $FA^+$ cation and its entropic contribution to the phase energetics were found to have a crucial role in stabilizing the cubic phase (Fig. 3C) (*12*, *60*), but the entropic contribution of the different A cations has barely been explored. Also, the effects of strain on OLHP thermodynamics and energetics have not been fully elucidated and require further systematic investigations. Defect states must also be correlated with the thermodynamic energetics of metastable OLHPs, which has been shown to affect the phase degradation of $FAPbI_3$ (*61*).

### *Impeding ion migration by steric and electrostatic interactions*

The charged ionic constituents of OLHPs are mobile under an electric field and accelerated under illumination or at increased temperatures. An ion displaced from its original lattice site becomes a crystal defect (or Schottky, Frenkel defect pairs). Ion (or defect) migration are known to cause the current-voltage hysteresis, phase segregation, and chemical corrosion of reactive functional layers, which are responsible for PSC degradation during operation. Strategies to inhibit ion migration can be broadly categorized as reducing the density of mobile ions and/or increasing the activation energy barrier ($E_a$) for ions to migrate.

Relative to single-cation compositions, substitutional doping to create mixed-cation compositions is broadly observed to alleviate PSC hysteresis. In mixed-cation compositions, the local lattice mismatch caused by A cations with different sizes was reported to distort the ion-migration pathways (Fig. 4, A and B) (*62*). This steric impediment effect was shown to increase $E_a$ to suppress ion migration, which consequently improved the thermal and photostabilities of PSCs. Conversely, excessive lattice strain may detrimentally lower $E_a$, because ion redistribution constitutes a driving force to relieve residual strain (*51*, *63*). This is especially relevant for pure $CsPbI_3$ and $FAPbI_3$ owing to their extreme *t* values. The intrinsic strain in $FAPbI_3$ was also shown to promote the formation of Schottky vacancy defects

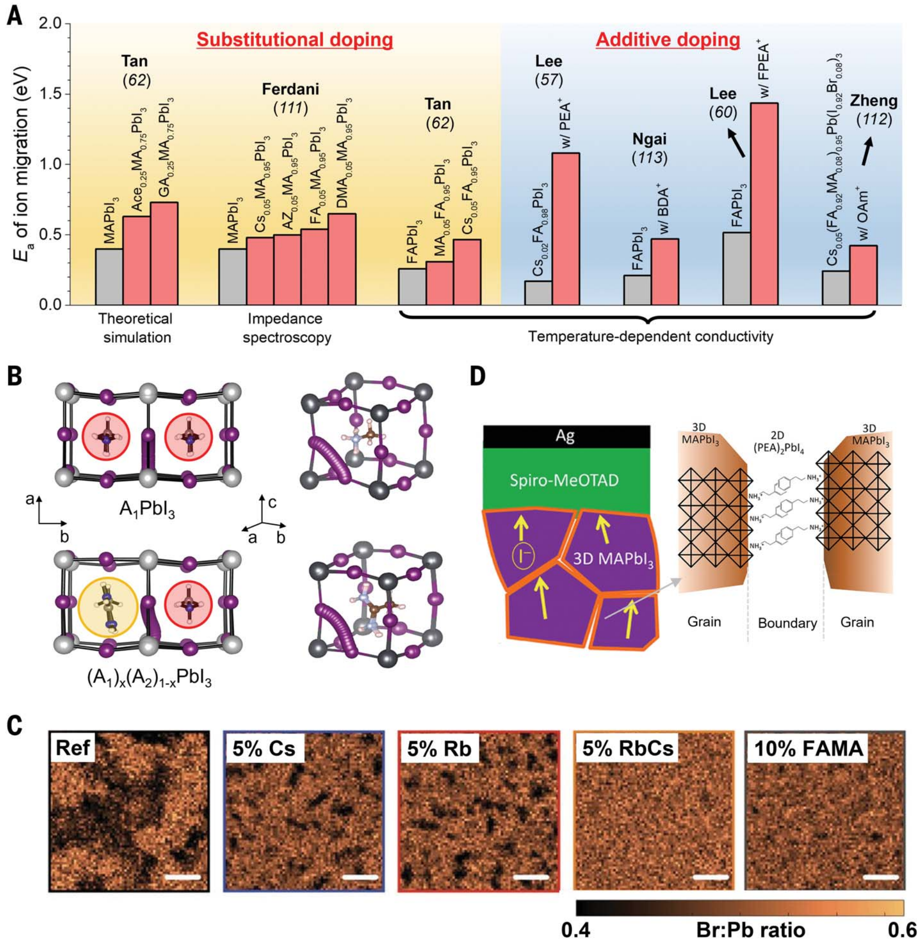


**Fig. 4. Impeding ion migration.** (**A**) Summary of reported $E_a$ enhancements by A cation engineering, including the measurement and calculation methodologies. (AZ, azetidinium; BDA, 1,4-butanediamonium; OAm, oleylaminium). The PCE data and compositions are retrieved from their respective publications (*57*, *60*, *62*, *111*–*113*). (**B**) Theoretically simulated iodide migration pathway for single-cation (top row) or mixed-cation (bottom row) compositions. The simulations were performed using the nudged elastic band and constrained energy minimization methodology. Adapted with permission from (*62*). (**C**) X-ray fluorescence mapping of the halide distribution of various mixed-halide perovskite films with different A cation compositions. Scale bars indicate 2 μm. Adapted with permission from (*65*). Ref, reference; FAMA, mixture of 8.3% FAI and 1.7% MAI. (**D**) Schematic illustrating the suppressed ion migration by grain boundary 2D $(PEA)_2PbI_4$. Adapted with permission from (*77*).

as a compensating strain relaxation mechanism (*64*). As a solution, partially substituting $FA^+$ in $FAPbI_3$ with smaller $Cs^+$ and/or $MA^+$ relaxes the intrinsic strain to suppress ion migration and defect formation (*41*, *51*). Such strain relaxation using $Cs^+$ has also been widely adopted to prevent phase segregation for wide-bandgap mixed-halide compositions (Fig. 4C) (*65*). Substitutional doping can also increase $E_a$ by strengthening the structural integrity of the OLHP lattice. This is achieved by increasing the bond number and/or decreasing the bond length between the A cation and the $BX_6^{4-}$ framework. This is illustrated by partial substitution of guanidinium ($GA^+$) into $MAPbI_3$, where $GA^+$ forms a greater number of effective hydrogen bonds than $MA^+$ ($GA^+$, seven bonds; $MA^+$, three bonds) with the $PbI_6^{4-}$ lattice (*62*, *66*). Molecular dynamics simulations have further indicated that ion migration is affected by the A cation polarity (*67*). The study suggests that more-polar A cations (e.g., $MA^+$ is more polar than $FA^+$) may detrimentally assist halide migration because of a reorientation and charge-screening mechanism.

The undersized alkali cations, focusing on $Rb^+$ and $K^+$, are widely used as additive dopants to reduce ion migration. The strongly electropositive cations can bind and immobilize excess and undercoordinated halides to suppress their unwanted migration (*68*). Theoretical simulations also indicate that alkali cation doping increases the formation energy of mobile halide interstitial defects (*69*). Additionally, assuming incorporation into the A site, the resultant lattice contraction decreases the ionic interaction distances to improve the structural integrity of the OLHP lattice (*43*, *46*). There is ongoing debate regarding the residing location of the undersized cations, especially for $Rb^+$. Experimental observations have been interpreted as incorporation into the lattice A site (*43*, *46*). Based on *t* values, their ionic radii should be too small to fit into the A site of the $APbI_3$ lattice. Theoretical calculations suggest that interstitial site occupation is energetically favorable (*70*–*72*). Conversely, solid-state NMR results suggest that $Rb^+$ and $K^+$ are entirely immiscible (even at the interstitial sites) and segregate into secondary photo-inactive phases (*73*). These discrepancies may perhaps be related to varying miscibility limits of different OLHP compositions. For example, $K^+$ incorporation may be facilitated by $Br^-$ in mixed-halide $APbI_{3-x}Br_x$ stoichiometries (*74*). Particularly, if assuming the undersized cations are indeed expelled from the A site, alternative mechanisms that do not involve lattice occupancy must be elucidated to explain their ion-migration impediment effects. For example, interstitial site occupancy has been suggested to prevent lattice iodine from displacing into the filled interstitial sites, which inhibits the formation of mobile iodide Frenkel defects (iodine interstitial-vacancy pairs) (*70*, *75*).

Oversized organic A cations are often used as additive dopants into the precursor solution. As discussed, the bulky cations segregate at the grain boundaries and surfaces, possibly forming lower-dimensional phases depending on the specific species. The bulky cations can act as physical barriers to increase $E_a$ by blocking the low-energy ion-migration channels through the grain boundaries (Fig. 4D) (*76*, *77*). The $–NH_3^+$ functional group of ammonium-based cations can also bind with negatively charged defects (e.g., A cation vacancies, undercoordinated halides) by electrostatic interactions to inhibit their migration and deactivate their charge-trapping ability. In an alternate perspective, the replacement of defective grain-boundary regions with lower-dimensional phases equivalently decreases the overall defect density, thus reducing the overall concentration of mobile defective species. Functional groups can be introduced to the A cation chain to strengthen the bonding interaction between adjacent cations along the grain boundary to improve structural rigidity. This is exemplified by incorporating a carboxyl tail into $BA^+$, forming 5-ammoniumvaleric acid (5-AVA). Adjacent 5-AVA cations interact through stronger hydrogen bonding rather than the van der Waals interaction between $BA^+$ cations. Resultantly, doping of 5-AVA into $MAPbI_3$ was reported to dramatically suppress the loss of MAI by ion migration–driven volatilization (*78*). Despite using the supposedly unstable $MAPbI_3$ base composition, carbon-based 10-cm–by–10-cm solar modules doped with $AVA^+$ were shown to sustain their performance for more than 10,000 hours under continuous illumination at 55°C (*79*).

Bulky cyclic cation additives, including 1-butyl-1-methylpiperidinium (BMP) and 1-butyl-3-methylimidazolium (BMIM), have been reported to dramatically inhibit ion migration and OLHP decomposition even at increased temperatures (*80*, *81*). It was proposed that BMP may simultaneously reduce the density of mobile iodide Frenkel defects and inhibit the migration of iodine interstitial defects, which quenched the formation rate of harmful $I_2$ that catalyzes perovskite degradation (*81*). Ion migration–facilitated formations of δ-$FAPbI_3$ and segregated $FAPbBr_3$ impurity phases during degradation may also be suppressed by BMP addition (*81*). Interestingly, BMIM was observed to aggregate predominantly at the buried interface, whereas BMP was distributed along the grain boundaries. It was suggested that the segregation of BMIM modified its interface dynamics to become incompatible with poly-TPD [poly(*N*,*N*′-bis-4-butylphenyl-*N*,*N*′-bisphenyl)benzidine] as the bottom transport layer, whereas BMP addition was not affected by this problem. This comparison highlights the fact that additive doping can possibly incur unintentional side effects to negatively alter the perovskite crystallization dynamics or interfacial energy alignments. More generally, this emphasizes the importance of locating the distribution of different additive dopants within and around the OLHP layer.

Ion migration is fundamentally a defect-driven process. Theoretical studies show that halide migration, mediated by vacancy defects, is the lowest-energy transport mechanism, with an $E_a$ between 0.1 and 0.6 eV (*82*). However, the vast majority of mitigation strategies in the literature fall short of pinpointing the specific defect species that are immobilized or suppressed. Further studies are necessary to investigate the binding and interaction of A cations with the complex defect landscape of OLHPs to mechanistically explain their ion-migration impediment effects.

### *Surface functionalization and interfacial modification*

The OLHP surface is typically $BX_2$-rich owing to a combination of volatilization of organic $FA^+$ or $MA^+$ during thermal annealing, dissolution of $FA^+$ or $MA^+$ by the deposition of solvent during surface posttreatment, or excess $BX_2$, particularly excess $PbI_2$, intentionally added to the precursor solution (*83*–*85*). Degradation initiation and nonradiative recombination losses predominantly occur at the OLHP top surface region as a result of an abundance of defective states (*61*, *86*). Consequently, surface modification by posttreatments using A cations has become an essential processing step for state-of-the-art PSCs in recent years (Fig. 5, A and B). Unlike additive doping into the precursor solution that was discussed previously, surface treatments do not interfere with the initial perovskite crystallization process, which enables a larger variety of candidate cations to be used. An additional secondary annealing step typically follows deposition, which may facilitate a chemical reaction with $BX_2$ to form thin lower-dimensional phases at the surface. The deposition of $PEA^+$ is a notable exception that omits the secondary annealing step. Formation of the 2D $(PEA)_2PbI_4$ was indeed observed to weaken the trap passivation effect (*87*). This distinction for different A cations, of whether formation of the lower-dimensional crystalline phase is preferred, is still not fully understood.

Surface functionalization of A cations often confers beneficial trap passivation and environmental protection effects. For the former, various functional groups on the cation can coordinate with surface defects to passivate their charge-trapping ability. Most commonly, the $–NH_3^+$ group of ubiquitous ammonium-based cations interacts with negatively charged defects (*85*). Chemical and structural tailoring of the cation backbone is often used to enhance its binding and interaction with surface traps. This is illustrated by the incorporation of the *tert*-butyl moiety (*25*, *88*), which also has the additional benefit of improving the interfacial contact with spiro-MeOTAD [2,2′,7,7′-tetrakis(4,4′-dimethoxy-3-methyldiphenylamino)-9,9′-spirobifluorene] to facilitate hole extraction (*88*). Conversely, the $–NH_3^+$ functionality is insensitive to positively charged defects, such as halide vacancies, which readily form at the OLHP surface. This can be remedied by using zwitterions to simultaneously coordinate with both oppositely charged defects, for instance, by incorporating carboxyl or phosphate groups into the ammonium chain (*25*, *89*). Regarding environmental protection, bulky A cations with hydrophobic chains that orient out of plane with the surface can repel $H_2O$ and $O_2$ ingression while simultaneously limiting the escape of the OLHP constituents from the active layer. Well-known hydrophobic groups include tertiary or quaternary hydrocarbons (e.g., tetra-ethylammonium) (*88*–*90*) and/or electron-withdrawing fluorine moieties, such as in 4-fluorophenylethylammonium or pentafluorophenylethylammonium (*60*, *91*). In general, a large variety of useful backbone-chain modifications can be found in the literature to enhance such desirable effects. However, despite these successes, concrete selection and design rules still need to be established to avoid

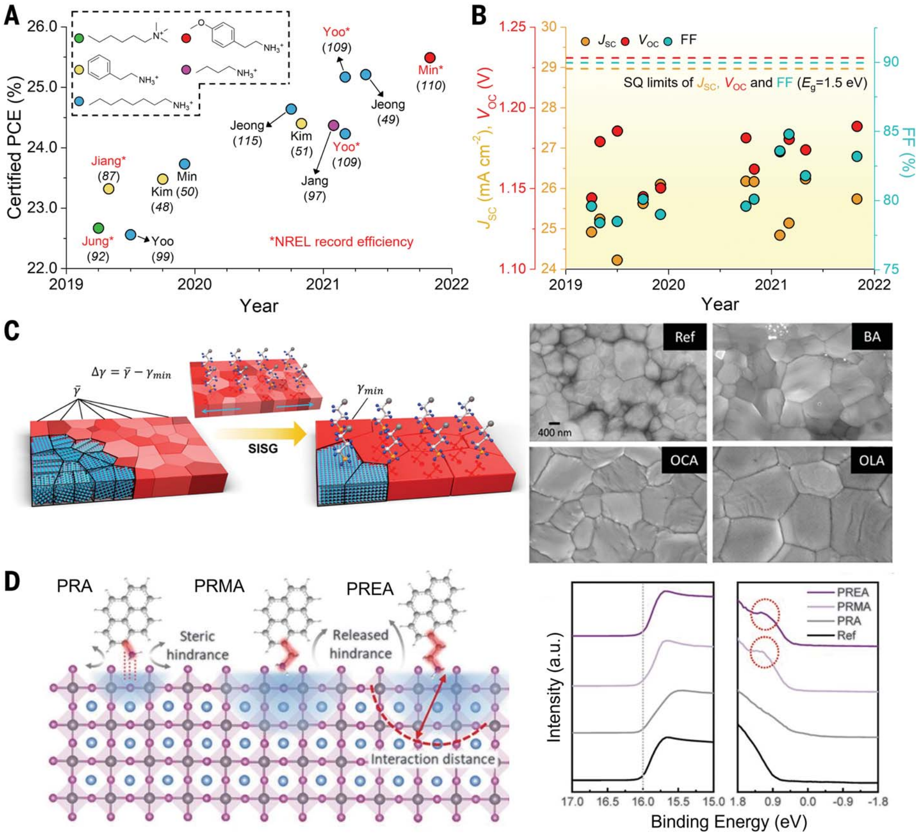


**Fig. 5. Surface and interface modification by oversized A cations.** (**A**) Summary of state-of-the-art PSCs with certified performance since 2019, highlighting the surface functionalization by the oversized A cations that were used. The data are plotted based on publication date. NREL, National Renewable Energy Laboratory. (**B**) Corresponding photovoltaic parameters from (A), with the horizontal dashed lines marking the theoretical Shockley-Queisser limits of each parameter for a bandgap of 1.5 eV. $J_{SC}$, short-circuit current; $V_{OC}$, open-circuit voltage; FF, fill factor. (**C**) Schematic of the surface-induced secondary grain growth (SISG) process by using alkylammonium functionalization to control the surface-energy anisotropy (left). Top-view scanning electron microscopy images of the perovskite films treated with various alkylammonium cations of different chain lengths (right). Adapted with permission from (93). γ, surface energy; OCA, octylammonium; OLA, oleylammonium. (**D**) Schematic illustrating the interactions of different pyrene-based ammonium cations with the OLHP film surface (left). Ultraviolet photoelectron spectroscopy spectra of OLHP films treated with the different cations (right). Red circles indicate the additional electronic states. Adapted with permission from (96). a.u., arbitrary units; PRA, pyrene-ammonium; PRMA, pyrene-methylammonium; PREA, pyrene-ethylammonium.

the utility of trial-and-error approaches and to maximize beneficial improvements.

In addition to their contributions to hydrophobicity and defect passivation, important breakthroughs in recent years have been realized by A cation surface modification. The top surface of OLHP films forms a carrier-extraction interface with a contacting charge-selective layer, typically a hydrophobic organic material. Surface modification by A cations can be used to aid the assembly of the charge-transporting layer and improve interfacial contact, by tailoring the interaction between the charge-transporting layer species and specific moieties on the A cation. This is exemplified by the use of *n*-hexyl-trimethylammonium ($HTA^+$) to facilitate the self-assembly of poly(3-hexylthiophene) (P3HT) (92). Interdigitation of the hydrophobic hexyl ($-C_6H_{13}$) backbone chains contained on both $HTA^+$ and P3HT promoted the growth of P3HT with a fibril structure. The fibril structure was suggested to enhance the charge-extraction capability of P3HT as a dopant-free hole-transporting material. This notable breakthrough resulted in one of the previous world-record performance PSCs (green dot in Fig. 5A). Relatedly, A cations can also be used to manipulate the OLHP surface energy. This can be exploited to induce the preferential growth of specific crystallographic planes with lower energy. During a secondary grain-growth process, $BA^+$, octylammonium, and oleylammonium progressively decreased the (100) plane surface energy to promote the recrystallization of highly textured (100) grains with greatly enlarged grain sizes (Fig. 5C) (93). Secondary surface grain growth has also been reported for A cations paired with $Br^-$ as the counter anion, including GABr (84), FABr (83), and MABr (94). Such grain recrystallization can effectively minimize surface pinhole and/or $PbI_2$ formation (84). More generally, rather than the ubiquitous $I^-$ (or $Br^-$), pairing A cations with alternative counter anions presents further opportunities to modify the OLHP surface, in addition to avoiding the possible formation of iodine interstitial defects as a side effect (61). For instance, sulfate ($SO_4^{2-}$) and phosphate ($PO_4^{3-}$) counter anions [paired with octylammonium ($OA^+$)] were reported to form thin (<5 nm) $PbSO_4$ or $Pb_3(PO_4)_2$ layers on the OLHP surface, which effectively inhibited ion migration owing to the strong chemical binding with the surface (95). Elsewhere, trifluoroacetate ($CF_3COO^-$) as a counter anion was reported to interact more strongly with halide vacancies on the surface (61).

As discussed, it was traditionally believed that A cations do not directly participate in constructing the OLHP band-edge electronic structure. This notion has since been challenged by the use of bulky polycyclic pyrene-based ammonium cations, where the frontier orbital separation of the cations was lowered by electron delocalization in the π-conjugated pyrene tails (Fig. 5D) (96). This enabled the orbitals of pyrene-methylammonium and pyrene-ethylammonium to overlap the I 5p and Pb 6s band-edge orbitals to contribute electronic states to the valence-band maximum, which increased hole mobility at the surface. Future works might explore whether the conjugated cations may have distorted the $PbX_6^{4-}$ octahedra differently, which could have contributed to altering the band dispersion.

Surface functionalization typically alters the OLHP energy-level alignment, which can beneficially promote carrier extraction by a contacting charge-selective layer. To maximize built-in potential, a pressure-assisted, solid-state growth strategy was reported by transfer from an as-grown $(BA)_2PbI_4$ template with annealing (97). The method was capable of growing a highly crystalline and phase-pure 2D $(BA)_2PbI_4$ layer on top of the 3D perovskite. Thickness control was demonstrated by simply changing the annealing temperature or by an iteration process. Conversely, different studies have reported the formation of either type I or type II band alignments for identical or similar A cation species (92, 97–100). Further mechanistic studies are necessary to explain the conflicting results. One possibility may be related to varying A cation distribution and orientation at the surface, which shift the OLHP band energetics differently (100). Both type I or type II alignments beneficially repel electrons away from a hole-selective interface, but type I alignments may represent an extraction barrier for holes. Relatedly, the chain groups of bulky A cations are typically electrically insulating. This effect manifests where, for example, with increasing alkylammonium chain length, the higher cation dipole moment enhances its interaction with traps to further reduce nonradiative recombination losses, but conversely, the longer insulating chain increases series resistance, which is detrimental for charge extraction (*99*, *101*). Likewise, it is often observed that with increasingly higher A cation surface density, the PSC open-circuit voltage typically increases monotonously but the fill factor peaks at an optimum concentration before subsequently decreasing as the insulating surface layer becomes too thick. It is therefore crucial to carefully design the A cation chain group and elucidate its density, distribution, and orientation on the OLHP surface to maximize its beneficial effects while ensuring that carrier extraction is not compromised.

## Perspectives

The structural flexibility of OLHPs has enabled A cation engineering to be a versatile approach to tailor and improve the properties of OLHPs for different applications. Future opportunities can be generally categorized as discovering new uses for existing A cation species or identifying new A cations for existing applications.

The former category necessitates a more comprehensive understanding of the structure-interaction-property-performance-stability correlations conferred by A cations. A fundamental understanding of the atomistic and mechanistic origins of reported beneficial improvements conferred by A cation engineering is still lacking. This includes understanding the physicochemical interactions of A cations with the lattice constituents, band structure, and charge carriers, as well as their assembly, occupancy, and distribution within the OLHP layer. For instance, the correlation between the dynamic A cation motion and OLHP charge-carrier dynamics is still under debate. This understanding may enable the design of targeted A cation compositions for different device applications, such as solar cells, photodetectors, or light-emitting diodes. Also, the spontaneous expulsion and assembly of oversized cations along the surfaces and grain boundaries are poorly understood but crucial to understand because of the generally insulating nature of the organic side chains.

For the latter category, despite the increasing number of candidates of oversized and undersized cations, only a limited number of cations can be bulk-incorporated into the lattice A site, limiting compositional diversity. Considerable efforts during the early days of OLHP research were devoted to identifying cations for bulk occupancy, but these have essentially ceased. This is related to an absence of working selection and design rules. The ideal bulk A cation should be slightly smaller than $FA^+$ but free of the volatility issues of $MA^+$. Besides size considerations, chemical design principles also include polarity, stereochemistry, coordination, and bonding. The characteristic dynamic nature of organic A cations should also be taken into account. These criteria are also applicable to chain-group modification of bulky organic cations. Chemical design of the A cation must further be correlated with the structural and thermodynamic stability of the lattice and its binding and interaction with the ionic and defective constituents of OLHPs. Particularly, cation polarity has been investigated relatively less but profoundly influences ion migration (*67*), surface energy (*93*), trap passivation, and charge extraction (*99*, *101*). Machine-learning methodologies have been gaining traction in the community, which may prove useful for accelerated screening and discovery.

Relatedly, to date, ammonium-based species constitute most of the A cations applied to OLHPs. However, the protic (acidic) ammonium group is susceptible to deprotonation by hydrogen donation, rendering it chemically vulnerable to water and radical and basic species. Therefore, the environmental stability of ammonium species is inherently limited by their proticity, despite the presence of hydrophobic chain groups. In this regard, replacing ammoniums with their analogous, aprotic sulfonium cations may offer a solution because of their lower acidity. It will be necessary to understand the likely different formability and interactions of sulfonium cations with the OLHP constituents and processing solvents, in consideration of the different polarity between sulfoniums and their ammonium counterparts.

More generally, for OLHP optoelectronics to achieve market readiness, the challenge should now be focused on addressing their long-term operational instability issues, which remain far short of the acceptable lifetime guarantees of commercial products. Engineering of A cations has contributed immensely to the stabilization improvements to date, but further progress is required. Reducing the device-to-module upscaling gap of OLHP technology is another crucial hurdle, and it is necessary to investigate the application and adaptability of A cation engineering strategies at the module scale. It is also important to investigate the applicability of Pb-based A cation engineering strategies when extending toward Sn or Pb-Sn stoichiometries, which are attractive for tandem applications or to reduce toxic Pb usage. There is still much work left to be done, and it will be important to fully understand the role and contribution of the A cation.

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## ACKNOWLEDGMENTS

**Funding:** This work was supported by a Korea Institute of Energy Technology Evaluation and Planning (KETEP) grant funded by the Korean government (MOTIE) under award numbers 20213030010400 and 20214000000640 (J.-W.L.); by National Research Foundation of Korea grants NRF-2018R1A3B1052820 (S.I.S.) and NRF-2021R1A3B1076723 (N.-G.P.); and by the US Department of Energy's Office of Energy Efficiency and Renewable Energy (EERE) under the Solar Energy Technologies Office under award number DE-EE0008751 (S.T. and Y.Y.) **Competing interests:** The authors declare that they have no competing interests.

10.1126/science.abj1186